CM3

Applications to discrepancy and ε-approximations

Where we are

- Lecture 1: discrepancy; we *announced* that a random coloring of any set system has discrepancy $O(\sqrt{n \ln m})$ with probability $\geq 1 - 1/m$.
- Lecture 2: expectation, linearity, Markov. Markov uses only the mean and is far too weak to see exponentially small tails.

The setup: sums of independent coin flips

Throughout:

$$X_1, \ldots, X_n \text{ independent}, \qquad X_i \in \{0,1\}, \qquad p_i = \Pr[X_i = 1],$$

$$X = \sum_{i=1}^{n} X_i, \qquad \mu = \mathbb{E}[X] = \sum_{i=1}^{n} p_i.$$

- Examples: n biased coins; random samples; the ± 1 signs of a random coloring (an affine change $2X_i - 1$ of $0/1$ coins).
- Independence is essential — without it, the bounds below are simply false.

Chernoff bounds: the general statement

Theorem 1 (Chernoff bounds, multiplicative form)

For every $\delta > 0$,

$$\Pr\big[\, X \geq (1+\delta)\,\mu \,\big] \;\leq\; \Big(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\Big)^{\mu},$$

and for every $0 < \delta < 1$,

$$\Pr\big[\, X \leq (1-\delta)\,\mu \,\big] \;\leq\; \Big(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\Big)^{\mu}.$$

Proof omitted: it is the standard exponential-moment argument (Markov applied to e^{tX}). We use these bounds as a black box.

The right-hand sides are < 1 and decay *exponentially* in μ.

From the multiplicative form to usable bounds

Lemma 2 (Calculus inequalities)

For $\delta \geq 0$,

$$\delta - (1+\delta)\ln(1+\delta) \;\leq\; -\frac{\delta^2}{2+\delta},$$

for $0 \leq \delta \leq 1$ *the left-hand side is in fact* $\leq -\delta^2/3$, *and for* $0 \leq \delta < 1$,

$$-\delta - (1-\delta)\ln(1-\delta) \;\leq\; -\frac{\delta^2}{2}.$$

Sketch.

Both sides vanish at $\delta = 0$. The third inequality follows from the series expansion $(1-\delta)\ln(1-\delta) = -\delta + \delta^2/2 + \delta^3/6 + \cdots \geq -\delta + \delta^2/2$; the first two are verified by differentiating the difference of the two sides and checking that the derivative is ≤ 0 on the stated range. □

Chernoff bounds: the simplified versions

Corollary 3 (The forms we will actually use)

1. *For all* $\delta \geq 0$: $\Pr\big[X \geq (1+\delta)\mu\big] \;\leq\; \exp\Big(-\dfrac{\delta^2\mu}{2+\delta}\Big)$.
2. *For* $0 \leq \delta \leq 1$: $\Pr\big[X \geq (1+\delta)\mu\big] \;\leq\; e^{-\delta^2\mu/3}$.
3. *For* $0 \leq \delta < 1$: $\Pr\big[X \leq (1-\delta)\mu\big] \;\leq\; e^{-\delta^2\mu/2}$.

Two-sided version: for $0 \leq \delta \leq 1$,

$$\Pr\big[\,|X-\mu| \geq \delta\mu\,\big] \;\leq\; 2\,e^{-\delta^2\mu/3}.$$

All bounds depend on X only through μ — the individual p_i do not matter.

How to read these bounds

$$\Pr\big[X \geq (1+\delta)\mu\big] \;\leq\; e^{-\delta^2\mu/3} \qquad (0 \leq \delta \leq 1)$$

- Exponential in μ: the number of independent trials is the currency. Twice as many trials $\Rightarrow$ the bound is *squared*.
- Quadratic in δ: doubling the relative deviation δ raises the bound to its *fourth* power: $e^{-(2\delta)^2\mu/3} = \left(e^{-\delta^2\mu/3}\right)^4$ — much smaller.
- Uniform over the p_i: only the sum $\mu = \sum_i p_i$ matters, however different the individual coins are.
- Two regimes: for $\delta \leq 1$ use $e^{-\delta^2\mu/3}$; for larger δ use the general form $e^{-\delta^2\mu/(2+\delta)}$ (valid for *all* $\delta \geq 0$).

Special case: fair coins

$X \sim \mathrm{Bin}(n, \frac{1}{2})$, so $\mu = n/2$. With $\delta = \frac{1}{2}$:

$$\Pr\left[X \geq \frac{3n}{4}\right] \;=\; \Pr\left[X \geq \left(1+\tfrac{1}{2}\right)\tfrac{n}{2}\right] \;\leq\; e^{-(1/4)(n/2)/3} \;=\; e^{-n/24}.$$

$$\Pr\left[X \leq \frac{n}{4}\right] \;=\; \Pr\left[X \leq \left(1-\tfrac{1}{2}\right)\tfrac{n}{2}\right] \;\leq\; e^{-(1/4)(n/2)/2} \;=\; e^{-n/16}.$$

(The general form for the upper tail gives the sharper $e^{-n/20}$.)

The tool for discrepancy: random signs

Corollary 4 (Hoeffding-type bound for ± 1 sums)

Let $Z_1, \ldots, Z_s$ be independent, each equal to $+1$ or -1 with probability $1/2$, and let $D = \sum_{i=1}^{s} Z_i$. Then for all $t \geq 0$,

$$\Pr\big[\,|D| \geq t\,\big] \;\leq\; 2\,e^{-t^2/(2s)}.$$

Hoeffding-type bound; proof omitted (same exponential-moment method as the Chernoff bounds).

Recall: Markov's inequality

Theorem 5 (Markov)

If $X \geq 0$ and $a > 0$, then

$$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}.$$

- Uses only the mean $\mathbb{E}[X]$ — nothing else.
- No independence, no variance, no bounds on the values (beyond nonnegativity).
- Price to pay: it is very weak. That is exactly what the comparison below shows.

The example: n fair coins, event $\{X \geq 3n/4\}$

Let $X \sim \mathrm{Bin}(n, \frac{1}{2})$, so $\mu = n/2$.

Markov:

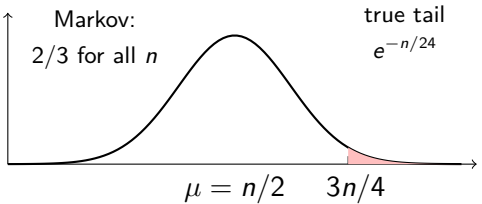
$$\Pr\left[X \geq \frac{3n}{4}\right] \leq \frac{n/2}{3n/4} = \frac{2}{3} \qquad \text{— for every } n!$$

Chernoff ($\delta = 1/2$):

$$\Pr\left[X \geq \frac{3n}{4}\right] \leq e^{-n/24}.$$

The gap explodes with n

n	10	100	1000	10,000
Markov $(2/3)$	0.667	0.667	0.667	0.667
Chernoff $(e^{-n/24})$	0.659	0.0155	$8 \cdot 10^{-19}$	10^{-181}



The true tail probability is exponentially small; Markov cannot see it.

Why is Markov so much weaker?

- Markov uses only the mean. It is *optimal* given only the mean: if X takes value a with probability p and 0 otherwise, then

$$\mathbb{E}[X] = ap, \qquad \Pr[X \geq a] = p = \frac{\mathbb{E}[X]}{a}$$

  — equality in Markov.
- To beat Markov we must exploit more structure. Here, the structure is *independence*: X is a sum of n independent coins, and Chernoff bounds exploit exactly that.
- Moral: the mean alone cannot certify concentration; independence can.

Check yourself

Question 1. We toss $n = 10^6$ fair coins; let X be the number of heads. Bound $\Pr[X \geq 550{,}000]$ with (a) Markov, (b) Chernoff.

(a) Markov: $\Pr[X \geq 550{,}000] \leq \frac{500{,}000}{550{,}000} \approx 0.909$ — useless.

(b) $\mu = 500{,}000$, $\delta = 0.1$:

$$\Pr[X \geq (1+0.1)\mu] \leq e^{-\delta^2\mu/3} = e^{-(0.01)(5\cdot 10^5)/3} = e^{-1667} \approx 10^{-724}.$$

Back to discrepancy (Lecture 1, recall)

- Set system $(X, \mathscr{R})$: X finite, $\mathscr{R}$ a set of subsets of X. Let $n = |X|$ and $m = |\mathscr{R}| \geq 2$.
- A two-coloring is $\chi : X \to \{-1, +1\}$; its discrepancy on $S \subseteq X$ is

$$\mathrm{disc}_\chi(S) = \left|\sum_{v \in S} \chi(v)\right|, \qquad \mathrm{disc}_\chi(\mathscr{R}) = \max_{S \in \mathscr{R}} \mathrm{disc}_\chi(S).$$

Goal: prove, with Chernoff bounds, that a random coloring has $\mathrm{disc}_\chi(\mathscr{R}) = O(\sqrt{n \ln m})$ with probability $\geq 1 - 1/m$.

Step 1: one fixed set behaves well

Fix $S \in \mathscr{R}$, write $s = |S|$, and color randomly:

$$D_S := \sum_{v \in S} \chi(v)$$

is a sum of s independent random signs. By the Hoeffding-type bound for ± 1 sums, for every $t \geq 0$:

$$\Pr\big[\, |D_S| \geq t \,\big] \;\leq\; 2\, e^{-t^2/(2s)}.$$

Take $t = \sqrt{2s \ln(2m)}$:

$$\Pr\Big[\, |D_S| \geq \sqrt{2s \ln(2m)} \,\Big] \;\leq\; 2\, e^{-\ln(2m)} \;=\; \frac{1}{m}.$$

Per-set bound: with probability $\geq 1 - 1/m$, $\operatorname{disc}_\chi(S) \leq \sqrt{2s \ln(2m)} = O\big(\sqrt{|S| \ln m}\big)$.

Step 2: all sets at once (union bound)

Union bound: for any events $E_1, \ldots, E_m$,

$$\Pr\big[\, E_1 \cup \cdots \cup E_m \,\big] \;\le\; \Pr[E_1] + \cdots + \Pr[E_m].$$

Now fix one common threshold $t > 0$, take $E_S = \{|D_S| \ge t\}$, and use $s = |S| \le n$:

$$\begin{aligned}\Pr\Big[\max_{S \in \mathscr{R}} |D_S| \ge t\Big] \;&\le\; \sum_{S \in \mathscr{R}} \Pr\big[\, |D_S| \ge t \,\big] \\ &\le\; \sum_{S \in \mathscr{R}} 2\, e^{-t^2/(2s)} \;\le\; 2m\, e^{-t^2/(2n)}.\end{aligned}$$

One bound per set $\Rightarrow$ one bound for the maximum, at the cost of a factor m.

Step 3: choosing the threshold

We want $2m\,e^{-t^2/(2n)} \le 1/m$. Take

$$t \;=\; 3\sqrt{n \ln m}.$$

Then $t^2 = 9n \ln m$, so

$$2m\,e^{-t^2/(2n)} \;\le\; 2m\,e^{-(9/2)\ln m} \;=\; 2m \cdot m^{-9/2} \;=\; 2\,m^{-7/2} \;\le\; \frac{1}{m} \qquad (m \ge 2).$$

Lemma 6 (Random coloring)

Let $(X, \mathscr{R})$ be a finite set system with $n = |X|$ and $m = |\mathscr{R}| \ge 2$, and let χ be a random two-coloring of X. Then, with probability at least $1 - 1/m$,

$$\operatorname{disc}_\chi(\mathscr{R}) \;=\; O\big(\sqrt{n \ln m}\big).$$

What the random coloring lemma gives

The lemma gives, in fact, *two* things:

1. there *exists* a coloring with discrepancy $O(\sqrt{n \ln m})$;
2. there is a simple randomized algorithm finding one with probability $\geq 1 - 1/m$: color randomly.

Remark 7

Out of the 2^n colorings of X, at least $(1 - 1/m)\, 2^n$ have discrepancy $O(\sqrt{n \ln m})$ — a huge fraction!

(The challenge of finding one *deterministically* is the subject of the next lectures.)

The bound in context

- Universal: $O(\sqrt{n \ln m})$ for *any* set system, with no structural assumptions — only n and m enter the proof.
- Special systems can be much better:
  - intervals on a line: discrepancy ≤ 1 (alternating coloring, Lecture 1);
  - halfplanes in the plane: optimal discrepancy $\Theta(n^{1/4})$.
- Special systems can be as bad as the bound suggests: systems from Hadamard matrices force discrepancy $\Omega(\sqrt{n})$.

The random bound is within a $\sqrt{\ln m}$ factor of optimal: Spencer's theorem gives discrepancy $O\left(\sqrt{n \ln(2m/n)}\right)$ for every system with $m \geq n$, and the Hadamard example above shows that $\Theta(\sqrt{n})$ is optimal when $m = n$.

Check yourself

Question 2. In the discrepancy proof: (a) which single inequality converts the per-set bound into a bound on $\max_{S \in \mathcal{R}} |D_S|$? (b) why is the choice $t = 3\sqrt{n \ln m}$ (rather than $t = \sqrt{2n \ln(2m)}$) convenient?

(a) The union bound: the probability that *some* set is bad is at most the sum of the individual bad probabilities.

(b) It makes the arithmetic clean: $2m\, e^{-t^2/(2n)} \le 2m^{-7/2} \le 1/m$, a success probability of $1 - 1/m$ with no fussy constants. In general $t = c\sqrt{n \ln m}$ works provided $c^2 \ge 4 + 2 \log_m 2$ (i.e. $c^2 \ge 2\big(2 + \log_m 2\big)$); the worst case is $m = 2$, so $c \ge \sqrt{6} \approx 2.45$ always works — our $c = 3$ is safely above this. (Note: $c = 2$ would give only $2/m$, not $1/m$.)

Motivation: compressing a set system

- $X = n$ points in the plane; $\mathscr{R}$ = the subsets induced by halfplanes.
- Query: *how many points of X lie in a given halfplane?*
- Storing all n points is expensive. Can a small sample A answer *every* query $S \in \mathscr{R}$ up to a small additive error?

Idea: replace X by a tiny multiset A such that, for every S,

$$\frac{|S \cap A|}{|A|} \approx \frac{|S|}{|X|}.$$

(Applications: range counting, statistics and polling, databases, computational geometry.)

ε-approximations: the definition

Definition 8 (ε-approximation)

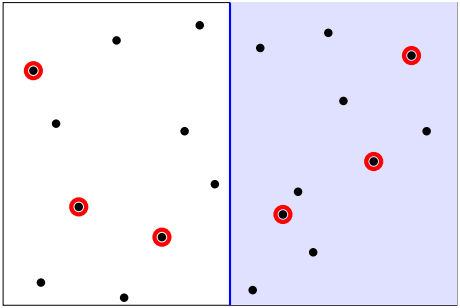
Let $(X, \mathscr{R})$ be a finite set system with n elements and m sets. Given a parameter $\varepsilon > 0$, an *ε-approximation* of $(X, \mathscr{R})$ is a multiset A over X satisfying

$$\text{for all } S \in \mathscr{R}: \qquad |S \cap A| \;=\; \frac{|S|\,|A|}{|X|} \;\pm\; \varepsilon\,|A|,$$

where $|S \cap A|$ denotes the number of elements of A lying in S, counted with multiplicity.

- In words: for *every* set S, the fraction of A inside S differs from the fraction of X inside S by at most ε.
- A is a multiset: elements may be sampled several times.

Intuition: what a random sample achieves



$|S|/|X| = 10/20 = 1/2$; the sample: $|S \cap A|/|A| = 3/6 = 1/2$

Picking the $|A|$ elements of A independently and uniformly at random (with replacement) from X: for each $S \in \mathscr{R}$,

$$\mathbb{E}\big[\,|S \cap A|\,\big] \;=\; \frac{|S|\,|A|}{|X|}.$$

The theorem: a random sample is an ε-approximation

Theorem 9 (Random sampling)

Let $(X, \mathscr{R})$ be a finite set system with n elements and m sets, and let $\varepsilon \in (0, 1/2]$. A uniform random sample A of size

$$a = \left\lceil \frac{3 \ln(2m^2)}{\varepsilon^2} \right\rceil = O\left(\frac{\ln m}{\varepsilon^2}\right)$$

(independent draws from X, with replacement) is an ε-approximation of $\mathscr{R}$ with probability at least $1 - 1/m$.

The size depends only on m and ε — not on n!

Proof, step 1: fix one set

Fix $S \in \mathscr{R}$, and write $p = |S|/|X|$. Let

$$X_S \;:=\; |S \cap A|$$

count how many of the a sampled elements lie in S. Each draw lands in S with probability p, independently, so

$$X_S \;\sim\; \mathrm{Bin}(a, p), \qquad \mu \;=\; \mathbb{E}[X_S] \;=\; ap.$$

Our goal for this set:

$$\Pr\big[\, |X_S - ap| \geq \varepsilon a \,\big] \;\leq\; \frac{1}{m^2}.$$

(Then a union bound over the m sets will finish the proof.)

Proof, step 2: the upper tail (simplified Chernoff)

If $p = 0$ the claim is trivial, so assume $p > 0$. Write $\varepsilon a = \delta\, ap$ with $\delta = \varepsilon/p$. The upper tail form

$$\Pr\big[X \geq (1+\delta)\mu\big] \;\leq\; \exp\Big(-\frac{\delta^2 \mu}{2+\delta}\Big) \qquad \text{(valid for } \textit{all}\ \delta \geq 0\text{)}$$

gives

$$\Pr\big[X_S \geq ap + \varepsilon a\big] \;\leq\; e^{-\varepsilon^2 a/(2p+\varepsilon)}.$$

Since $p \leq 1$ and $\varepsilon \leq 1$:

$$2p + \varepsilon \;\leq\; 3 \qquad \Longrightarrow \qquad \Pr\big[X_S \geq ap + \varepsilon a\big] \;\leq\; e^{-\varepsilon^2 a/3}.$$

The $(2+\delta)$ form is exactly what makes this uniform in p — no case analysis needed.

Proof, step 3: the lower tail

- If $p \le \varepsilon$, then $ap - \varepsilon a \le 0$ and the event $\{X_S \le ap - \varepsilon a\}$ is empty — nothing to prove.
- If $p > \varepsilon$, set $\delta = \varepsilon/p < 1$ and use the lower tail form $\Pr[X \le (1-\delta)\mu] \le e^{-\delta^2\mu/2}$:

$$\Pr\big[X_S \le ap - \varepsilon a\big] \;\le\; e^{-\varepsilon^2 a/(2p)} \;\le\; e^{-\varepsilon^2 a/2} \;\le\; e^{-\varepsilon^2 a/3}.$$

Combining both tails:

$$\Pr\big[\,|X_S - ap| \ge \varepsilon a\,\big] \;\le\; e^{-\varepsilon^2 a/3} + e^{-\varepsilon^2 a/3} \;=\; 2\,e^{-\varepsilon^2 a/3}.$$

Proof, step 4: union bound over the m sets

With $a = \lceil 3\ln(2m^2)/\varepsilon^2 \rceil$:

$$2\,e^{-\varepsilon^2 a/3} \;\le\; 2\,e^{-\ln(2m^2)} \;=\; \frac{1}{m^2}.$$

Union bound over all $S \in \mathscr{R}$:

$$\Pr\big[\,A \text{ is not an } \varepsilon\text{-approximation}\,\big] \;\le\; \sum_{S\in\mathscr{R}} \Pr\big[|X_S - ap| \ge \varepsilon a\big] \;\le\; m \cdot \frac{1}{m^2} \;=\; \frac{1}{m}.$$

Hence, with probability at least $1 - 1/m$, A is an ε-approximation of $\mathscr{R}$. □

Why is the sample size independent of n?

- For *one* fixed set S, the error probability depends only on the sample size a and $p = |S|/|X| \leq 1$ — never on n.
- The $\ln m$ term is the price of asking about *all* m sets at once (union bound).
- So we may compress X from n elements down to $O(\ln m/\varepsilon^2)$, however large n is!

Remark 10 (Deterministic version, next lecture)

There is an iterative (multiplicative weights) algorithm computing an ε-approximation of size $O(\ln m/\varepsilon^2)$ — the same bound, with no randomness at all.

Discrepancy and ε-approximations are siblings

Proposition 11 (A low-discrepancy coloring gives an ε-approximation)

Let n be even and let $\chi : X \to \{-1, +1\}$ be a two-coloring with $\operatorname{disc}_\chi(\mathscr{R}) = \Delta$ and exactly $n/2$ vertices of each color. Then $A = \chi^{-1}(+1)$ has $|A| = n/2$ and is an ε-approximation of $(X, \mathscr{R})$ with $\varepsilon = \Delta/n$.

Proof.

For $S \in \mathscr{R}$: since $|A|/|X| = 1/2$,

$$\left| |S \cap A| - \frac{|S|}{2} \right| = \left| \frac{1}{2} \sum_{v \in S} \chi(v) \right| \leq \frac{\Delta}{2} = \frac{\Delta}{n} \cdot \frac{n}{2} = \varepsilon |A|.$$

□

Balancing colors and compressing data are the same activity: a random balanced coloring has $\Delta = O(\sqrt{n \ln m})$ w.h.p., hence gives an ε-approximation of size $n/2$ with $\varepsilon = O(\sqrt{\ln m / n})$.

Check yourself

Question 3. We want an ε-approximation with $\varepsilon = 0.1$ for a set system with $m = 1000$ sets. (a) How many random samples suffice (with the constants of the proof)? (b) Why does the answer not depend on $n = |X|$?

(a) $a = \left\lceil \frac{3 \ln(2m^2)}{\varepsilon^2} \right\rceil = \left\lceil \frac{3 \ln(2 \cdot 10^6)}{0.01} \right\rceil \approx \lceil 4353 \rceil = 4353$ samples.

(b) The per-set error probability decays only in the sample size a; n appears nowhere in the Chernoff exponent. m enters through the union bound as $\ln m$.